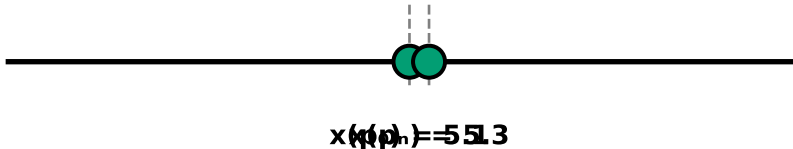


FP32 Floating-Point (Works)

Activation values



CDC Gradient (element-wise difference):

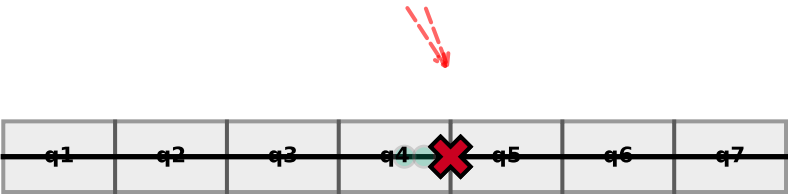
$$\Delta = x(p_0 + p_n) - x(p_0)$$

$$\Delta = 5.3 - 5.1 = 0.2$$

✓ *Non-zero gradient*
→ *Backprop works*

INT8 Quantized (Fails)

Quantization buckets



CDC Gradient (after INT8 quantization):

$$\Delta_q = q(x(p_0 + p_n)) - q(x(p_0))$$

$$\Delta_q = q5 - q5 = 0$$

✗ *Zero gradient*
→ *Backprop blocked*
→ *Model collapse*

INT8 quantization destroys CDC's micro-gradient structure